Exercise (Recitation Sep 24)**.** Consider the function

$$f(x) = \int_0^1 \frac{1 - e^{-xt^2}}{t}\, dt.$$

1. Find the set D of x for which $f(x)$ is well-defined and real-valued.
2. Prove that $f \in C^1(D)$.
3. Compute $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$.
4. Prove that the function
$$F(x, y) = \int_0^1 \frac{1 - e^{-xyt^2}}{t}\, dt$$
is C^1 in an open set of $\mathbb{R}^2$.
5. Compute
$$\lim_{(x,y)\to(0,0)} \frac{F(x, y)}{\sqrt{x^2 + y^2}}.$$

Proof. 1. $D = \mathbb{R}$. To show this, fix x and let

$$g(t) = \frac{1 - e^{-xt^2}}{t}.$$

First show that $g(t) \to 0$ as $t \to 0$. With Taylor's theorem, we have

$$e^{-xt^2} = 1 - xt^2 + o(t^3).$$

Thus $1 - e^{-xt^2} = xt^2 + o(t^3)$. Note that the little o here is considering the limit as $t \to 0$. Therefore, $g(t) \to 0$ as $t \to 0$.

Since g is continuous on $(0, 1]$, it is measureable. Now if $g(t) \to 0$ as $t \to 0$, we can show $f(x) < \infty$ for any $x \in \mathbb{R}$. Indeed, by the limit above, there exists $0 < \delta < 1$ such that $g(t) < 1$ on $(0, \delta]$. Since g continuous $g(t)$ is bounded on compact interval $[\delta, 1]$. Hence, $g(t)$ is bounded on $(0, 1]$, and thus $f(x) < \infty$ for all $x \in \mathbb{R}$.

2. First show that the derivative of f is

$$T(x) = \int_0^1 te^{-xt^2}\, dt.$$

This is well defined, as for each fixed x, we have

$$te^{-xt^2} \le \max\{e^{-x}, 1\}$$

for $t \in [0, 1]$. Therefore, $T(x) < \infty$.

Fix $x \in \mathbb{R}$. We want to show

$$\lim_{u \to 0} \frac{|f(x+u) - f(x) - T(x)u|}{|u|} = 0.$$

Let $\{u_n\}_{n=1}^{\infty}$ be any sequence with $u_n \to 0$ and $|u_n| \leq 1$. We have

$$\frac{|f(x+u_n) - f(x) - T(x)u_n|}{|u_n|} \leq \frac{1}{|u_n|} \int_0^1 \left| \frac{e^{-xt^2} - e^{-(x+u_n)t^2}}{t} - te^{-xt^2}u_n \right| dt.$$

By mean value theorem we have

$$\left| \frac{e^{-xt^2} - e^{-(x+u_n)t^2}}{t} - te^{-xt^2}u_n \right| \leq |u_n| \sup_{\xi \in L(0,u_n)} \left| te^{-(x+\xi)t^2} - te^{-xt^2} \right|.$$

Then,

$$\frac{|f(x+u_n) - f(x) - T(x)u_n|}{|u_n|} \leq \int_0^1 \sup_{\xi \in L(0,u_n)} \left| te^{-(x+\xi)t^2} - te^{-xt^2} \right| dt.$$

Towards using DCT, note that

$$\left| te^{-(x+\xi)t^2} - te^{-xt^2} \right| \leq 2t \max\{e^{-x+1}, 1\} \leq 2 \max\{e^{-x+1}, 1\}$$

for $t \in [0,1]$. Note that the RHS is a constant w.r.t. t and thus integrable on $[0,1]$. By DCT, this implies

$$\begin{aligned}
\lim_{n \to \infty} \frac{|f(x+u_n) - f(x) - T(x)u_n|}{|u_n|} &\leq \lim_{n \to \infty} \int_0^1 \sup_{\xi \in L(0,u_n)} \left| te^{-(x+\xi)t^2} - te^{-xt^2} \right| dt \\
&= \int_0^1 \lim_{n \to \infty} \sup_{\xi \in L(0,u_n)} \left| te^{-(x+\xi)t^2} - te^{-xt^2} \right| dt \\
&= 0,
\end{aligned}$$

since $u_n \to 0$ as $n \to \infty$, $\xi \in [0, u_n]$, and $\xi \mapsto te^{-(x+\xi)t^2}$ is continuous.

Since $\{u_n\}_{n=1}^{\infty}$ is arbitrary sequence converges to 0, this shows the desired limit and $Df = T$.

To show $f \in C^1(\mathbb{R})$, it remains to show T is continuous, i.e. $T(x+u) - T(x) \to 0$ as $u \to 0$. For this again let $\{u_n\}_{n=1}^{\infty}$ be any sequence with $u_n \to 0$ and $|u_n| \leq 1$. We have

$$T(x+u_n) - T(x) \leq \int_0^1 \left| te^{-(x+u_n)t^2} - te^{-xt^2} \right| dt.$$

Again note

$$\left| te^{-(x+u_n)t^2} - te^{-xt^2} \right| \leq 2t \max\{e^{-x+1}, 1\} \leq 2 \max\{e^{-x+1}, 1\}$$

for $t \in [0,1]$, which is integrable on $[0,1]$. Therefore, by DCT,

$$\begin{aligned}\lim_{n\to\infty} T(x+u_n) - T(x) &\le \lim_{n\to\infty} \int_0^1 \left| te^{-(x+u_n)t^2} - te^{-xt^2} \right| dt \\ &= \int_0^1 \lim_{n\to\infty} \left| te^{-(x+u_n)t^2} - te^{-xt^2} \right| dt \\ &= 0,\end{aligned}$$

since $u_n \to 0$ as $n \to 0$, and $u \mapsto te^{-(x+u)t}$ is continuous.

Since $\{u_n\}_{n=1}^{\infty}$ is arbitrary sequence converges to 0, this shows the desired limit and completes the proof.

3. Claim that $\lim_{x\to\infty} f(x) = \infty$. Let $\{x_n\}_{n=1}^{\infty}$ be a sequence such that $x_n \ge 0$ and $x_n \to \infty$. We then have

$$\frac{1-e^{-xt^2}}{t} \ge 0.$$

It then follows from MCT that

$$\lim_{n\to\infty} \int_0^1 \frac{1-e^{-x_n t^2}}{t}\, dt = \int_0^1 \lim_{n\to\infty} \frac{1-e^{-x_n t^2}}{t}\, dt = \int_0^1 \frac{1}{t}\, dt = \infty.$$

Since $\{x_n\}_{n=1}^{\infty}$ is an arbitrary sequence, this shows that $\lim_{x\to\infty} f(x) = \infty$.

Similarly, it easily follows from MCT that $\lim_{x\to-\infty} f(x) = -\infty$.

4. $F \in C^1(\mathbb{R}^2)$. Similar to part 1, we have

$$g(x,y) = \frac{1-e^{-xyt^2}}{t} \to 0 \text{ as } t \to 0,$$

so $F(x,y) < \infty$ for all $(x,y) \in \mathbb{R}^2$. To show $F \in C^1(\mathbb{R}^2)$, we first show

$$DF(x,y)(h,k) = \int_0^1 Dg(x,y)(h,k)\, dt,$$

where

$$Dg(x,y)(h,k) = (yh+xk)e^{-xyt^2}.$$

Fix $(x,y) \in \mathbb{R}^2$ and denote the linear map on RHS w.r.t. (h,k) as T. Let $\{(h_n,k_n)\}_{n=1}^{\infty}$ be a sequence with $(h_n,k_n) \to 0$, and consider

$$\begin{aligned}&\frac{|f(x+h_n,y+k_n) - f(x) - T(h_n,k_n)|}{\|(h_n,k_n)\|} \\ &= \frac{1}{\|(h_n,k_n)\|} \left| \int_0^1 g(x+h_n,y+k_n) - g(x,y) - Dg(x,y)(h_n,k_n)\, dt \right|\end{aligned}$$

Towards using DCT, mean value theorem give

$$|g(x+h_n, y+k_n) - g(x,y) - Dg(x,y)(h_n,k_n)| \le \sup_{z\in L} \|Dg(z) - Dg(x,y)\| \, \|(h_n,k_n)\|,$$

where the supremum is taken over the line segment L from (x,y) to $(x+h_n, y+k_n)$. Therefore

$$\frac{|f(x+h_n,y+k_n) - f(x) - T(h_n,k_n)|}{\|(h_n,k_n)\|} \le \int_0^1 \sup_{z\in L} \|Dg(z) - Dg(x,y)\| \, dt$$

Now we have

$$(Dg(z) - Dg(x,y))(h,k) = (z_2 h + z_1 k)e^{-z_1 z_2 t^2} - (yh + xk)e^{-xyt^2}.$$

To find $\|Dg(z) - Dg(x,y)\|$, let (h,k) be such that $\|(h,k)\| \le 1$. When n is large enough, we have $|(z_2 - y)h + (z_1 - x)k| \le 1$, $|z_2 h + z_1 k| \le |xh + yk| + 1$, and $|z_1 z_2| \le |xy| + 1$. It follows that for all $0 \le t \le 1$, we have

$$(Dg(z) - Dg(x,y))(h,k) \le (|xh+yk|+1)e^{|xy|t^2+1} \le (|x|+|y|+1)e^{|xy|t^2+1},$$

which is integrable w.r.t. t on $[0,1]$. Therefore, DCT applies and gives $DF(x,y)(h,k) = \int_0^1 Dg(x,y)(h,k)\,dt$. The proof for DF being continuous is similar.

5. The limit is zero. By Taylor's theorem, there exists $\delta > 0$ such that when $xyt^2 < \delta$,

$$\left|1 - e^{-xyt^2} - xyt^2\right| \le \left|xyt^2\right|.$$

It follows that

$$\left|\frac{1 - e^{-xyt^2}}{\sqrt{x^2+y^2}}\right| \le \frac{2\,|xy|\,t^2}{\sqrt{x^2+y^2}} \le 2t^2$$

for all $0 \le t \le 1$ when (x,y) are sufficiently close to $(0,0)$. Now let $\{(x_n,y_n)\}_{n=1}^\infty$ be a sequence such that $(x_n,y_n) \to (0,0)$. We then have

$$\int_0^1 \left|\frac{1 - e^{-x_n y_n t^2}}{t\sqrt{x_n^2+y_n^2}}\right| dt \le \int_0^1 2t\,dt < \infty.$$

It then follows from DCT that

$$\lim_{n\to\infty} \frac{F(x_n,y_n)}{\sqrt{x_n^2+y_n^2}} = \int_0^1 \lim_{n\to\infty} \frac{1 - e^{-x_n y_n t^2}}{t\sqrt{x_n^2+y_n^2}}\,dt = 0.$$

Since $\{(x_n,y_n)\}_{n=1}^\infty$ is an arbitrary sequence, this implies that

$$\lim_{(x,y)\to(0,0)} \frac{F(x,y)}{\sqrt{x^2+y^2}} = 0.$$

□

Remark. Some useful forms of mean value theorem: Let X and Y be normed vector space and $U \subset X$ open. Suppose that $x, y \in U$ with $x \neq y$ and that

$$L(x,y) = \{\alpha y + (1-\alpha x) : \alpha \in [0,1]\} \subset U.$$

Suppose that $f : U \to Y$ is continuous on $L(x,y)$ and differentiable on $L(x,y) \setminus E$, where $E \subset L(x,y)$ is countable. Then the following holds:

1. If
$$\sup_{z \in L(x,y)\setminus E} \|Df(z)\| = M < \infty,$$
then
$$\|f(y) - f(x)\| \leq M \|x - y\|.$$
2. If $T \in \mathcal{L}(X;Y)$ and
$$\sup_{z \in L(x,y)\setminus E} \|Df(z) - T\| = K < \infty,$$
then
$$\|f(y) - f(x) - T(y-x)\| \leq K \|x - y\|.$$

Exercise (Midterm 1)**.** Let

$$f_n(x) = nx^2(1-x)^n \cos(2\pi x), \quad x \in [0,1].$$

1. Does the series of functions $\sum_{n=1}^{\infty} f_n$ converges pointwise on $[0,1]$.
2. Does the equality
$$\sum_{n=1}^{\infty} \int_0^1 f_n \, dx = \int_0^1 \sum_{n=1}^{\infty} f_n \, dx$$
hold?
3. Does the series of function $\sum_{n=1}^{\infty} f_n$ converges uniformly on $[0,1]$?

Proof. 1. Yes. The series clearly converges at 0 and 1. For $0 < x < 1$, we have

$$|f_n(x)| \leq nx^2(1-x)^n.$$

Notice that $0 < 1 - x < 1$, so the series converges by root test.

2. Yes. Notice that

$$\int_0^1 |f_n| \leq \int_0^1 nx^2(1-x)^n = \frac{2n}{(n+1)(n+2)(n+3)}.$$

Therefore,

$$\int_0^1 \left| \sum_{n=1}^{\infty} f_n \right| \leq \int_0^1 \sum_{n=1}^{\infty} |f_n| = \sum_{n=1}^{\infty} \int_0^1 |f_n| < \infty$$

The desired equality then follows from DCT.

3. No. For $N, M \in \mathbb{N}$ and $M \geq N$, consider

$$\sum_{n=N}^{M} f_n(x) = \sum_{n=N}^{M} nx^2(1-x)^n \cos(2\pi x).$$

Note that $x \mapsto x^2(1-x)^n$ attains its maximum at $\frac{2}{2+n}$, so consider

$$\begin{aligned}
\left\| \sum_{n=N}^{M} f_n \right\| &\geq \sum_{n=N}^{M} f_n \left(\frac{2}{2+n} \right) \\
&= \sum_{n=N}^{M} \frac{4n}{(n+2)^2} \left(\frac{n}{n+2} \right)^n \cos\left(\frac{4\pi}{n+2} \right) \\
&\geq \sum_{n=N}^{M} \frac{n}{(n+2)^2} \frac{1}{e^2}
\end{aligned}$$

for N large enough. However note that $\sum_{n=1}^{\infty} \frac{n}{(n+2)^2}$ diverges, so the partial sums is not Cauchy. It follows that $\sum_{n=1}^{\infty} f_n$ is not uniformly Cauchy either, so the series of function does not converge uniformly on $[0, 1]$.

□